

Maxwell–Wagner Relaxation of the Capacitive Memory Window in Hafnia Ferroelectric Capacitors

Apu Das, Agniva Paul, Gautham Kumar, Debraj Barman, Ankita Dutta, Yannick Raffel, and Sourav De

Abstract—The capacitive memory window (CMW) enables nondestructive readout in ferroelectric capacitors, yet the mechanism governing its frequency and temperature limits remains unclear. Here, we show that polarization defines the stored states, whereas an interfacial charge layer governs their zero-bias capacitive discrimination. In TaN/HZO/TiN capacitors, the CMW follows a single-Debye relaxation with a corner frequency of $f_c = 179$ kHz. Independent analyses of the real and imaginary dielectric responses agree within 2%, and no nonrelaxing contribution is resolved on the intrinsic ferroelectric timescale. At 10 K, f_c decreases below 1 kHz, although polarization switching remains observable. Two device stacks with nearly identical low-frequency dielectric ceilings exhibit a $126\times$ difference in relaxation time. Series-capacitance analysis reveals only a 26% difference in interfacial areal capacitance but a $157\times$ difference in areal resistance, identifying interfacial conductivity, rather than interfacial thickness, as the dominant determinant of read speed. These results support an interface-limited Maxwell–Wagner relaxation mechanism and show that single-frequency CMW values cannot be directly compared across different device stacks.

Index Terms—Ferroelectric capacitor, hafnium zirconium oxide, capacitive memory window, nondestructive readout, Maxwell–Wagner relaxation.

I. INTRODUCTION

FERROELECTRICITY in doped hafnium-oxide (HfO₂), especially in Hf_{0.5}Zr_{0.5}O₂ (HZO) has enabled scalable ferroelectric devices compatible with complementary metal-oxide-semiconductor (CMOS) and back-end-of-line processing [1]–[4]. HZO retains polarization at nanometer-scale thickness and can be deposited conformally by atomic layer deposition, making ferroelectric capacitors (FeCaps) and ferroelectric field-effect transistors (FeFETs) promising for embedded non-volatile memory, storage-class memory, and in-memory and neuromorphic computing [5]–[8].

Conventional FeCap readout is destructive because the stored state is identified by switching the polarization through

the coercive field. The cell must subsequently be rewritten, increasing read energy, latency, and endurance consumption. Capacitive readout avoids this operation by sensing the small-signal capacitance under a subcoercive excitation. The difference between the capacitances of the two remanent polarization states defines the capacitive memory window (CMW), enabling nondestructive readout and capacitance-based in-memory computation [9], [10].

Two approaches have been used to obtain a measurable CMW. In asymmetric metal–ferroelectric–metal stacks, dissimilar interfaces generate a finite window at zero read bias. Such devices have demonstrated CMW values approaching 8.7 and read endurance beyond 10^{11} cycles, with subsequent implementations employing IGZO interlayers and three-dimensional architectures [9]. Alternatively, symmetric stacks produce little or no zero-bias CMW and are read using a finite subcoercive voltage, thereby reducing concerns associated with built-in asymmetry and device-to-device variability [9]. Despite these advances, CMW is commonly reported at a single frequency, and the physical mechanism governing its bandwidth and temperature dependence remains unclear.

Polarization is required to define the two nonvolatile states, but it is not sufficient to produce a zero-bias capacitive window. In an ideally symmetric capacitor, the small-signal permittivity is invariant under polarization reversal,

$$\xi(+P_r) = \xi(-P_r), \quad (1)$$

and the two C – V branches coincide at zero bias regardless of the magnitude of $2P_r$. A finite zero-bias CMW therefore requires an additional asymmetry that distinguishes the two polarization states. Determining the origin and dynamics of this asymmetry is essential for defining the usable read-frequency range and the stability of the capacitive read margin.

The electrode/HZO interface is a likely source of this behavior. TiN and TaN can scavenge oxygen during deposition and crystallization, producing oxygen-deficient TiO_xN_y and TaO_xN_y regions and redistributing oxygen vacancies near the HZO interfaces [11]–[13]. Interface and bulk resistance components differing by several orders of magnitude have been resolved by impedance spectroscopy in HZO capacitors [14], [15]. Interfacial defects and oxygen-vacancy redistribution have also been associated with wake-up, imprint, retention, and endurance degradation [3], [11], [16]–[18]. Direct structural and electrical studies further show that vacancy distribution, phase evolution, and interface-trap accumulation strongly influence the reliability of HZO capacitors and FeFETs [12], [13], [15], [17], [18]. Correspondingly,

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engineered $\text{WO}_x/\text{HZO}/\text{PEALD-TiN}$ interfaces have enabled FeCap endurance beyond 2×10^{10} cycles [9].

These observations motivate two distinct interpretations of the capacitive window. If the CMW is governed primarily by the intrinsic ferroelectric response, its relaxation should follow the high-frequency dielectric and domain-switching dynamics of HZO, remain comparatively insensitive to the electrode stack, and exhibit broadening associated with distributed domain or defect energies [2], [19]–[22]. In contrast, an interface-controlled CMW should exhibit a characteristic relaxation time that depends strongly on the interfacial impedance and shifts markedly with electrode configuration and temperature.

In this work, we distinguish between these mechanisms using frequency-resolved C – V and conductance spectroscopy from 1 kHz to 4 MHz on two HZO capacitor stacks with different electrode interfaces. Polarization switching is independently evaluated using positive-up-negative-down measurements and switching-current analysis at 300 and 10 K. The combined frequency, stack, and temperature dependence shows that polarization defines the stored states, whereas the bandwidth of their zero-bias capacitive discrimination is governed by an interfacial Maxwell–Wagner relaxation.

II. EXPERIMENTAL METHODS

Two metal–ferroelectric–metal capacitors with different electrode configurations were investigated. Stack A consisted of $\text{TaN}/\text{HZO}/\text{TiN}/\text{Si}$, whereas Stack B consisted of $\text{TiN}/\text{HZO}/\text{WO}_x/\text{W}/\text{Si}$. The HZO layers were deposited by atomic layer deposition and crystallized by rapid thermal annealing below 400°C with the top electrode present. Crystallization and stack-dependent diffraction features were assessed by grazing-incidence X-ray diffraction. Oxygen-deficient interfacial regions are expected at the nitride/HZO contacts because of oxygen scavenging, while Stack B additionally contains a deposited WO_x interfacial layer.

Small-signal C – V measurements were performed in parallel C_p – R_p mode using a 50-mV-rms excitation at 61 logarithmically spaced frequencies from 1 kHz to 4 MHz. The dc voltage was swept as $-2 \rightarrow +2 \rightarrow -2$ V to record both polarization branches under identical measurement conditions. The effective relative permittivity of each device was calculated as

$$\xi_r(V, f) = \frac{C_p(V, f)d}{\varepsilon_0 A} = \frac{C_p(V, f)}{C_0}, \quad C_0 = \frac{\varepsilon_0 A}{d}, \quad (2)$$

where d is the HZO thickness, A is the capacitor area, and C_0 is the corresponding geometric capacitance. For Stack A, $C_0 = 15.0$ pF.

The zero-bias capacitive memory window was defined as

$$\text{CMW}_\xi(f) = \xi_\downarrow(0, f) - \xi_\uparrow(0, f), \quad (3)$$

where $\xi_\downarrow$ and $\xi_\uparrow$ denote the branches reached after positive and negative poling, respectively. The crossing voltage, V_{cross} , was determined from

$$\xi_\downarrow(V_{\text{cross}}, f) = \xi_\uparrow(V_{\text{cross}}, f), \quad (4)$$

and the corresponding crossing field was calculated as $E_{\text{cross}} = V_{\text{cross}}/d$. The differential dielectric loss was obtained independently from the measured conductance, $G = 1/R_p$, as

$$\Delta\xi''(f) = \frac{G_\downarrow(0, f) - G_\uparrow(0, f)}{\omega C_0}, \quad \omega = 2\pi f. \quad (5)$$

High-frequency fixture and electrode contributions were evaluated using a series R_s – L – C equivalent circuit fitted over 0.8–4 MHz. The extracted parameters were $R_s = 267 \Omega$, $L = 14 \mu\text{H}$, and $C = 329$ pF. To minimize parasitic-circuit contributions, the principal dielectric analysis was restricted to $f \leq 500$ kHz.

Polarization was evaluated using positive-up-negative-down (PUND) measurements. The positive and negative switched charges were determined from

$$Q_{\text{sw}}^+ = \int [I_P(t) - I_U(t)] dt, \quad Q_{\text{sw}}^- = \int [I_N(t) - I_D(t)] dt, \quad (6)$$

where P and N denote switching pulses and U and D denote the corresponding nonswitching pulses. The switchable polarization was then calculated by averaging the magnitudes of the positive and negative switched charges:

$$2P_r = \frac{|Q_{\text{sw}}^+| + |Q_{\text{sw}}^-|}{2A}. \quad (7)$$

Switching-current and P – E measurements were performed using triangular waveforms with a half-period of $500 \mu\text{s}$. Switching kinetics were characterized using rectangular write pulses of ± 2 V with widths ranging from $10 \mu\text{s}$ to 1 ms. The positive and negative coercive voltages, V_c^+ and V_c^- , were obtained from the zero crossings of the P – V branches. The imprint voltage and coercive field were calculated as

$$V_{\text{imp}} = \frac{V_c^+ + V_c^-}{2}, \quad E_c = \frac{|V_c^+| + |V_c^-|}{2d}. \quad (8)$$

Cryogenic measurements were performed at 10 K under vacuum after a minimum stabilization time of 30 min, with the stage temperature maintained within ± 0.1 K. Identical probes, cabling, and instrument settings were used for the 300- and 10-K measurements. For the CMW analysis, Debye fitting was restricted to $f \leq 500$ kHz for Stack A and $f \leq 20$ kHz for Stack B. Peak-permittivity fitting was limited to $f \leq 300$ and 30 kHz for Stacks A and B, respectively. The leakage-corrected Cole–Cole response of Stack A was fitted over 30–300 kHz.

III. RESULTS AND DISCUSSION

A. Structural Characterization of the Capacitor Stacks

Figure 1 compares the cross-sectional structures and grazing-incidence X-ray diffraction (GIXRD) patterns of the two capacitor stacks. Stack A consists of a 107-nm TaN top electrode, a 9.49-nm HZO layer, and a 44.57-nm TiN bottom electrode on Si [Fig. 1(a)]. Stack B contains a 25-nm TiN top electrode, a 9.49-nm HZO layer, and a 25-nm W bottom electrode [Fig. 1(b)]. In both structures, the HZO layer retains nearly identical physical thickness, allowing the electrical differences between the two devices to be examined without a substantial change in ferroelectric thickness.

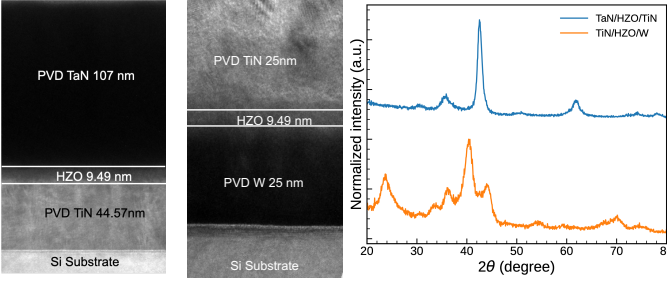


Fig. 1. Cross-sectional micrographs and normalized GIXRD patterns of the two ferroelectric capacitor stacks. Stack A had the nominal structure TaN(120 nm)/HZO(10 nm)/TiN(50 nm)/Si. Cross-sectional micrograph analysis yielded corresponding physical thicknesses of approximately 107, 9.49, and 44.57 nm, respectively. Stack B consists of TiN(25 nm)/HZO(9.49 nm)/W(25 nm)/Si. The WO_x interfacial layer in Stack B is not separately resolved at this magnification. The diffraction patterns are vertically offset for clarity.

The nominal WO_x interfacial region in Stack B is not separately resolved at the scale of the cross-sectional micrograph.

The corresponding GIXRD patterns are shown in Fig. 1(c). The two stacks exhibit distinctly different diffraction signatures because of their dissimilar electrode materials and textures. The TaN/HZO/TiN stack is dominated by strong nitride-related reflections, whereas the TiN/HZO/W stack contains additional W-related contributions. Since the HZO film is only approximately 9.5 nm thick and several HZO reflections overlap with the electrode peaks, the full-range patterns are used primarily to establish crystallization and stack-dependent texture.

B. Capacitive Memory Window and Its Frequency Dependence

Figure 2 compares the zero-bias capacitive memory window and its frequency evolution in the two capacitor stacks. At 1 kHz, Stack A exhibits the characteristic butterfly-shaped permittivity response [Fig. 2a]. The two polarization branches remain separated at zero electric field, with $\xi_{\downarrow}(0) = 55.0$ and $\xi_{\uparrow}(0) = 49.8$, corresponding to $\text{CMW}_{\xi} \approx 5.2$. Stack B also exhibits a butterfly response with a comparable peak relative permittivity, but its zero-bias branch separation is smaller, yielding $\text{CMW}_{\xi} \approx 2.3$ [Fig. 2b]. The similar peak permittivity of the two stacks indicates that the difference in CMW does not originate simply from a lower dielectric response of the HZO layer.

The frequency-dependent measurements reveal that the CMW is associated with a dispersive component of the dielectric response. In Stack A [Fig. 2c], increasing the frequency from 1 to 400 kHz progressively reduces both the butterfly-peak amplitude and the separation between the two polarization branches. At 400 kHz, the branches are nearly coincident, although a substantial relative permittivity of approximately 27–30 remains. Therefore, the frequency increase suppresses the capacitive distinction between the two stored states without eliminating the underlying dielectric response.

Stack B displays the same qualitative behavior but at a substantially lower frequency scale [Fig. 2d]. Its polarization branches are already nearly merged throughout the 50–400 kHz range, consistent with a much lower relaxation corner.

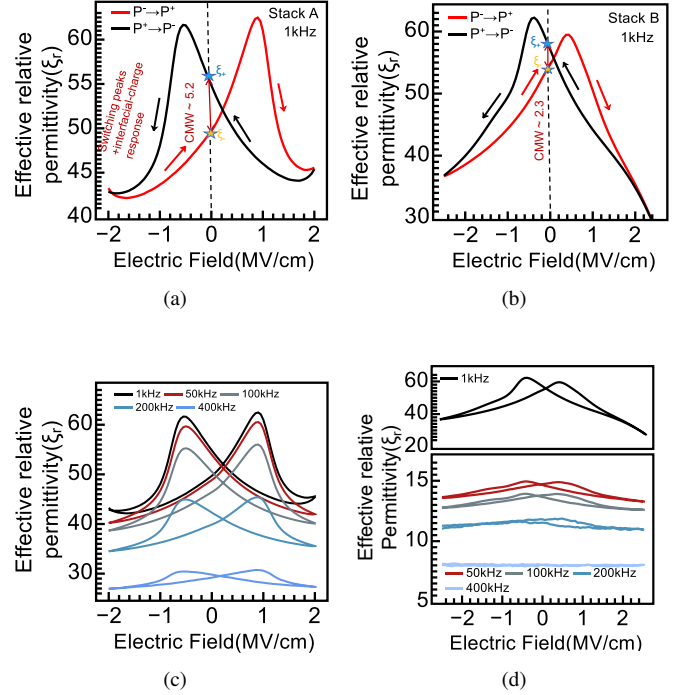


Fig. 2. Capacitive memory window and frequency-dependent dielectric response at 300 K. (a),(b) Effective relative permittivity of Stacks A and B at 1 kHz, respectively. The zero-field separation between the two polarization branches yields $\text{CMW}_{\xi} \approx 5.2$ for Stack A and $\text{CMW}_{\xi} \approx 2.3$ for Stack B, despite their comparable peak permittivity. (c),(d) Frequency-dependent permittivity of Stacks A and B, respectively. Increasing frequency progressively suppresses the branch separation while a finite state-independent dielectric response remains; the collapse occurs at a substantially lower frequency in Stack B.

These observations indicate that the measured response contains a state-dependent relaxing contribution superimposed on a comparatively nonrelaxing dielectric background. The similar dielectric ceilings but markedly different relaxation rates further suggest that the CMW dynamics are governed by the capacitor stack rather than by the HZO layer alone.

C. Interfacial Control of the Relaxation Clock

The markedly different frequency scales observed in Fig. 2 cannot be explained by the HZO dielectric response alone. Figure 3 examines whether the electrode/HZO interfaces instead govern the relaxation clock. The schematic in Fig. 3a illustrates the effective interfacial regions considered for the two stacks. Oxygen scavenging at the nitride contacts can produce oxygen-deficient TaO_xN_y and TiO_xN_y regions in Stack A, whereas Stack B additionally contains the deposited WO_x layer. These electrically active regions can support trapped and mobile charge and therefore contribute a series interfacial impedance.

Despite their different electrode configurations, the two stacks exhibit nearly identical low-frequency dielectric ceilings [Fig. 3c]. Debye extrapolation gives $\xi_r(f \rightarrow 0) = 62.4 \pm 1.2$ for Stack A and 62.7 ± 0.9 for Stack B, consistent with their measured 1-kHz peak permittivities. This similarity indicates that the large difference in the CMW relaxation rate does not originate from a substantially different HZO dielectric response.

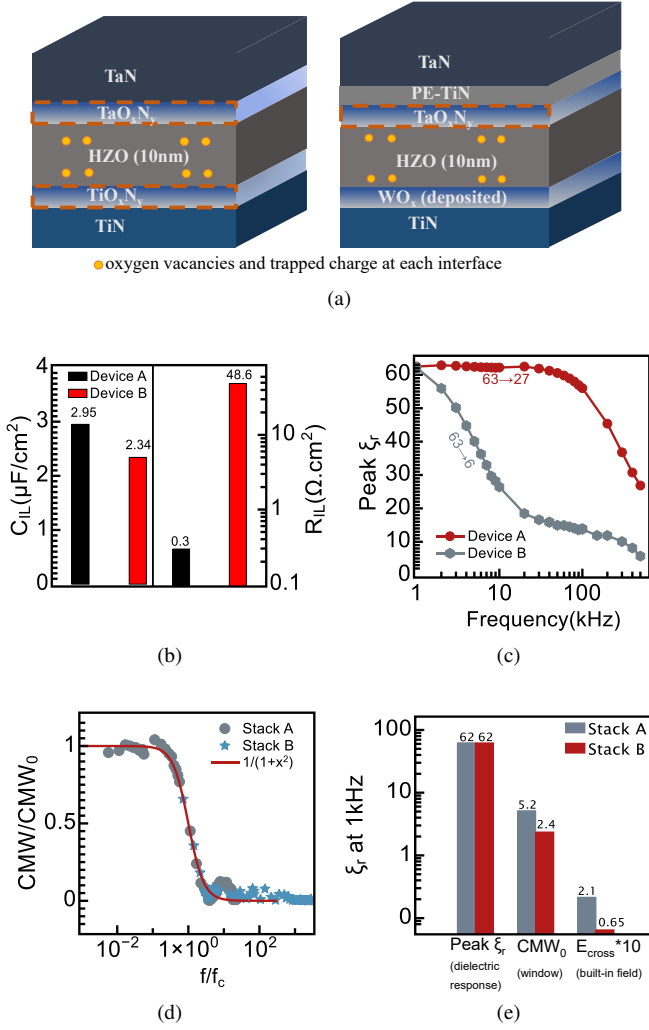


Fig. 3. Interfacial control of the capacitive memory-window relaxation. (a) Schematics of the two capacitor stacks including the effective oxygen-deficient interfacial regions at the nitride/HZO contacts and the deposited WO_x layer in Stack B. (b) Extracted interfacial areal capacitance and resistance. The two stacks exhibit comparable c_{IL} but differ by approximately 160 $\times$ in R_{IL} . (c) Peak relative permittivity versus frequency. Both stacks extrapolate to nearly identical low-frequency dielectric ceilings, while their high-frequency limits differ. (d) Normalized CMW versus f/f_c , showing collapse of both datasets onto the single-Debye response. (e) Comparison of the peak permittivity, CMW, and crossing-field metric for Stacks A and B.

In contrast, the fitted high-frequency limits are $\xi_\infty = 21.7 \pm 0.9$ and 15.8 ± 0.5 for Stacks A and B, respectively, consistent with different series interfacial impedances. The two CMW datasets differ strongly in characteristic time: $f_c = 179$ kHz and $\tau = 0.90$ μs for Stack A, compared with $f_c = 1.4$ kHz and $\tau = 114$ μs for Stack B. However, normalization by the fitted plateau and corner frequency collapses both datasets onto $1/[1 + (f/f_c)^2]$ [Fig. 3d]. Thus, the two devices exhibit the same functional relaxation but operate with substantially different clocks.

The effective interfacial parameters were extracted from the low- and high-frequency permittivity limits using $c_{IL} = \epsilon_0/[\epsilon_\infty - \epsilon_{FE}^{-1}]$ and $R_{IL} = \tau/c_{IL}$. As shown in Fig. 3b, Stacks A and B yield $c_{IL} = 2.95$ and 2.34 $\mu\text{F}/\text{cm}^2$, respectively, whereas their areal resistances are approximately 0.30 and 48.6 $\Omega\cdot\text{cm}^2$. The interfacial capacitances therefore differ

only modestly, while the resistance differs by approximately 160 $\times$. The much slower response of the WO_x-containing stack is therefore associated primarily with its larger effective interfacial resistance rather than with an increased series capacitance.

Figure 3e summarizes the comparison. The two stacks possess nearly identical peak dielectric responses but different CMW amplitudes, crossing fields, and relaxation times. The larger crossing field is accompanied by a larger zero-bias CMW, consistent with interfacial asymmetry setting the window amplitude; however, two stacks are insufficient to establish a universal scaling relation. Collectively, the results show that the HZO layer defines the dielectric ceiling, whereas the effective interfacial impedance controls the operating frequency of the capacitive readout. The extracted c_{IL} and R_{IL} represent the combined contribution of both interfaces and cannot be assigned to an individual interface from electrical measurements alone.

D. Single-Debye Relaxation of the Capacitive Memory Window

To determine whether the CMW originates from a single relaxation process or from a broad distribution of relaxation times, the two polarization states were represented by

$$\xi_\pm(\omega) = \xi_\infty + \frac{\Delta\xi_\pm}{1 + j\omega\tau}, \quad (9)$$

where ξ_∞ is the state-independent high-frequency permittivity, $\Delta\xi_\pm$ is the relaxing contribution, and τ is the characteristic relaxation time. Because ξ_∞ cancels when the two polarization states are subtracted, the frequency-dependent CMW becomes

$$\text{CMW}(f) = \frac{\text{CMW}_0}{1 + (f/f_c)^2}, \quad f_c = \frac{1}{2\pi\tau}. \quad (10)$$

Figure 4a shows the measured zero-bias CMW of both stacks together with the corresponding single-Debye fits. Stack A yields $\text{CMW}_0 = 5.38$ and $f_c = 179$ kHz, corresponding to $\tau = 0.90$ μs . In contrast, Stack B yields $\text{CMW}_0 = 3.6$ and $f_c = 1.4$ kHz, corresponding to $\tau = 114$ μs . The relaxation times therefore differ by approximately two orders of magnitude. For Stack B, the low-frequency plateau is obtained from the fit because the measured value at 1 kHz already lies on the relaxation roll-off.

An independent determination of the characteristic frequency was obtained from the conductance response. The differential dielectric loss was fitted using

$$\Delta\xi''(f) = \frac{A_{dc}}{f} + \text{CMW}_0 \frac{f/f_c}{1 + (f/f_c)^2}, \quad (11)$$

where the first term accounts for the state-dependent dc leakage contribution. As shown in Fig. 4b, the fit to Stack A gives a Debye loss maximum at 182 kHz, in close agreement with the 179-kHz corner obtained from the real-part response. Because the capacitance and conductance channels are measured independently, this agreement supports the physical origin of the extracted relaxation frequency.

Figure 4c presents the differential Cole–Cole response after removal of the leakage contribution. Data between 30 and

300 kHz were used for the circular-arc fit; lower-frequency points are dominated by the leakage subtraction, whereas higher-frequency points are affected by series resistance. The fitted depression angle is $\alpha = -2.7^\circ$, which is indistinguishable from zero within the experimental uncertainty. Constraining the response to an ideal Debye semicircle increases the rms radial residual only from 3.1% to 3.5%, indicating that a distributed relaxation-time model is not required. The Cole–Cole amplitude of 5.50 also agrees within 2.1% with $\text{CMW}_0 = 5.38$ obtained from the real-part fit.

The consistent characteristic frequencies obtained from the CMW roll-off and differential-loss maximum, together with the nearly undepressed Cole–Cole response, demonstrate that the CMW is governed by one dominant relaxation process. Furthermore, introducing a nonrelaxing CMW component produces an unphysical negative offset, limiting any unresolved high-frequency contribution to less than approximately 7% of CMW_0 . The measured corner is also well below the circuit-limited frequency of 1.81 MHz, confirming that the 179-kHz relaxation is intrinsic to the capacitor stack rather than to the measurement circuit.

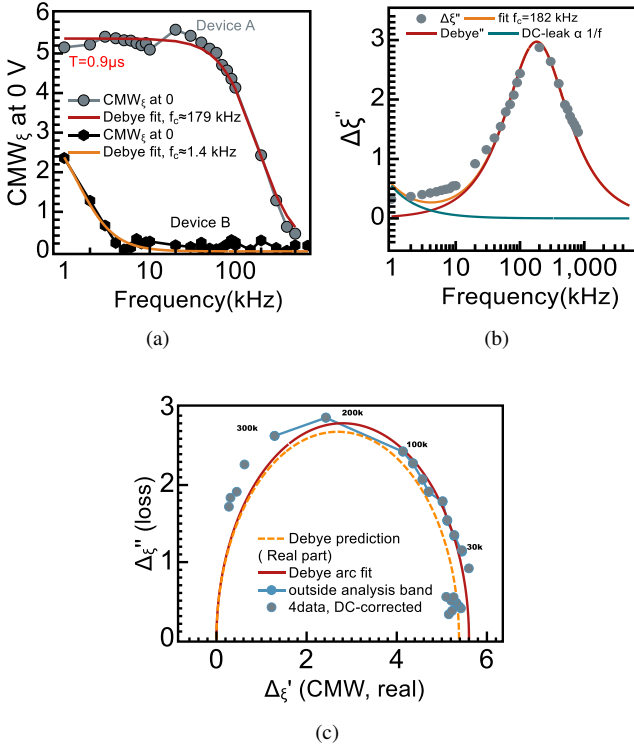


Fig. 4. Single-Debye relaxation of the capacitive memory window. (a) Zero-bias CMW of Stacks A and B with Debye fits, yielding $f_c = 179$ kHz ($\tau = 0.90$ μ s) and $f_c = 1.4$ kHz ($\tau = 114$ μ s), respectively. (b) Differential dielectric loss of Stack A, decomposed into a dc-leakage contribution proportional to $1/f$ and a Debye loss peak at 182 kHz. (c) Leakage-corrected Cole–Cole response over 30–300 kHz. The fitted depression angle of -2.7° is consistent with an ideal Debye response; the fitted amplitude of 5.50 agrees within 2.1% with $\text{CMW}_0 = 5.38$ obtained from the real-part fit. Open symbols denote data excluded from the fitting range.

E. Cryogenic Decoupling of Polarization Switching and Capacitive Readout

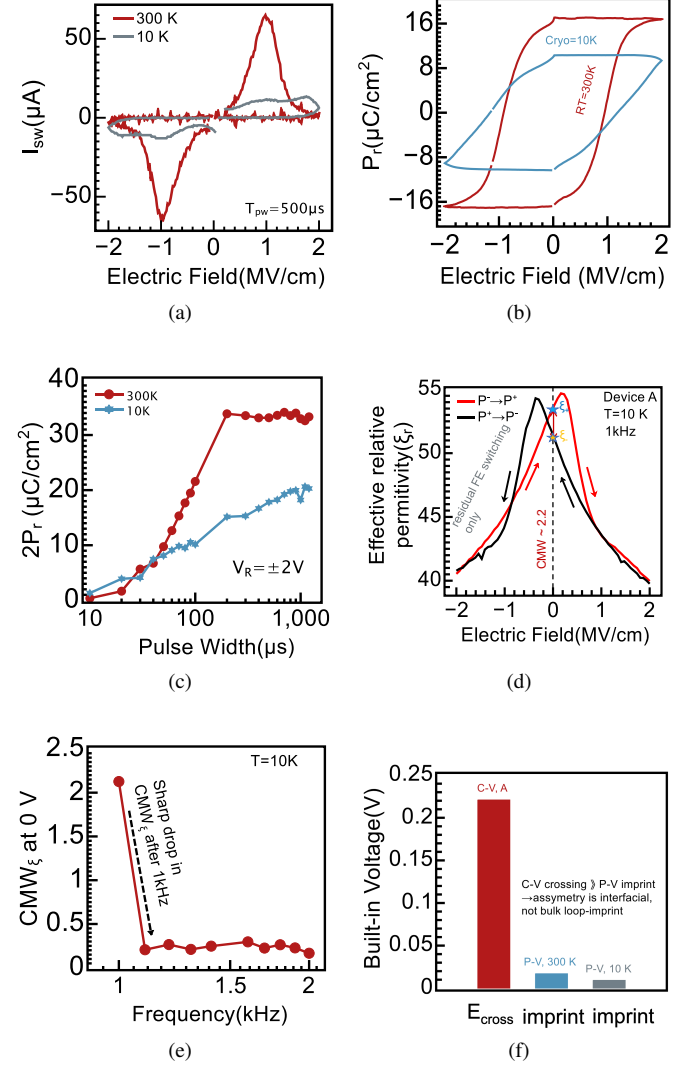


Fig. 5. Cryogenic evolution of polarization switching and capacitive readout in Stack A. (a) Switching-current characteristics at 300 and 10 K for $T_{pw} = 500$ μ s, showing reduced, broadened, and higher-field switching peaks at 10 K. (b) Corresponding $P-E$ loops; $2P_r$ decreases from 33.1 to 20.3 $\mu\text{C}/\text{cm}^2$, while E_c increases from 0.93 to 1.17 MV/cm. (c) Pulse-width dependence of $2P_r$ at $V_R = \pm 2$ V. The unsaturated response at 10 K indicates slower switching kinetics. (d) Butterfly $C-V$ response at 10 K and 1 kHz, retaining a zero-bias CMW of approximately 2.2. (e) Zero-bias CMW versus frequency at 10 K. The response approaches the noise floor above 1.1 kHz, placing an upper bound of $f_c \lesssim 1$ kHz on the relaxation corner. (f) Comparison of the $P-E$ loop imprint and $C-V$ crossing voltage. The substantially larger crossing voltage indicates that the asymmetry governing the capacitive window is distinct from the ferroelectric-loop imprint and is consistent with an interfacial contribution.

Figure 5 examines whether the capacitive memory window follows the ferroelectric polarization upon cooling. The switching-current characteristics of Stack A remain clearly resolved at 10 K [Fig. 5a], although the peak current decreases from approximately ± 65 μ A at 300 K to about 13 μ A and shifts toward higher electric field. The corresponding $P-E$ loops [Fig. 5b] show that the switchable polarization decreases from $2P_r = 33.1$ to 20.3 $\mu\text{C}/\text{cm}^2$, while the coercive field

increases from 0.93 to 1.17 MV/cm. Thus, cooling suppresses and slows the switching response but does not eliminate ferroelectric polarization reversal.

The pulse-width dependence in Fig. 5c shows that the reduced polarization at 10 K is primarily kinetic. At 300 K, $2P_r$ approaches $33 \mu\text{C}/\text{cm}^2$ by a pulse width of approximately $200 \mu\text{s}$. At 10 K, however, $2P_r$ continues to increase up to the longest applied pulse width of 1 ms and does not reach saturation within the measured range. Therefore, the smaller polarization obtained at a fixed pulse width reflects slower switching kinetics rather than the disappearance of the ferroelectric state.

The capacitive readout exhibits a stronger temperature dependence. The butterfly C - V response measured at 10 K and 1 kHz retains a finite zero-bias window of approximately 2.2 [Fig. 5d], compared with 5.2 at 300 K. However, the frequency-dependent measurement in Fig. 5e shows that the window approaches the experimental noise floor above 1.1 kHz. The available data therefore do not resolve the complete low-temperature relaxation spectrum but place an upper bound of approximately $f_c \lesssim 1$ kHz on its corner frequency. Cooling consequently shifts the capacitive-read bandwidth by more than two orders of magnitude, even though polarization switching remains measurable.

Figure 5f compares the asymmetry extracted from the ferroelectric hysteresis with that obtained from the capacitance response. The P - E loop imprint, $(V_c^+ + V_c^-)/2$, is only 0.017 V at 300 K and 0.009 V at 10 K, whereas the C - V crossing voltage is 0.221 V. The much larger crossing voltage shows that the asymmetry governing the zero-bias CMW is not equivalent to the imprint of the polarization loop. Together with the cryogenic loss of CMW bandwidth while switching persists, this distinction is consistent with an interfacial electrostatic contribution controlling the capacitive discrimination between the two polarization states. The data support this interpretation, although direct spatial identification of the responsible interfacial charge requires additional structural or spectroscopic characterization.

IV. CONCLUSION

The zero-bias capacitive memory window in HZO capacitors is governed by an interface-controlled relaxation rather than by the intrinsic ferroelectric response alone. Stack A exhibits a single-Debye relaxation with $f_c = 179$ kHz and $\tau = 0.90 \mu\text{s}$, independently confirmed by a 182-kHz differential-loss maximum and a nearly ideal Cole–Cole response. Stack B follows the same normalized behavior but with $f_c = 1.4$ kHz and $\tau = 114 \mu\text{s}$. Although both stacks show nearly identical low-frequency dielectric ceilings, their effective interfacial resistances differ by approximately two orders of magnitude, whereas their interfacial areal capacitances differ by only 26%. At 10 K, polarization switching remains measurable while the CMW corner shifts below 1 kHz. These results support an interface-limited Maxwell–Wagner mechanism and show that CMW values should be reported together with their relaxation corner frequency. Effective interfacial resistance is therefore a key design parameter for high-speed and temperature-stable capacitive readout.

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